

Final Project of FIT3139

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Section 1: Specification table

Base model	Schelling's Segregation Model An agent-based model where individuals move based on a threshold preference for neighbours of the same type, leading to unintended segregation
Extension	1. Location Preferences: Agents move based on neighbour similarity and the closeness to "desired locations" 2. Stochastic Relocation: Agents randomly relocate to model real-life unpredictability
Technique 1	Markov Chain: Agent happiness is evaluated deterministically as the system's transitions follow a Markov-like process. Unhappy agents move randomly to empty spaces, and each step depends on the grid's present state, with no memory of past states.
Technique 2	Monte Carlo Simulation: Approximate steady-state segregation outcomes by iteratively random sampling agent movements under stochastic conditions.
Modelling questions 1	How does introducing location preferences change the clustering dynamics shown in the original Schelling model?
Modelling questions 2	What circumstances cause emergent segregation patterns to be disrupted or reinforced by stochastic relocation?

Section 2: Introduction

Problem & Motivation

The Schelling model shows how personal choices of each individual can lead to widespread segregation without intentional discrimination. Actual residential systems are more complicated; people are randomly reassigned due to employment changes or other interruptions, which influence them to take into account the proximity of their preferred location. This extension of location preference and random movement explores whether segregation persists and results in clustering.

Extension Overview

Location Preferences

Agents consider their happiness based on proximity to certain "desirable locations" such as a central park, school, or shopping district. The closer an agent is to these zones, the happier they are overall with the effect weighted by a proximity weight (α), a spatial utility component. As a result, agents will cluster based on similarity with neighbours and distance proximity to "desirable locations".

Stochastic Relocation

Any agent, regardless of their happiness satisfaction rule, has a fixed probability p of moving to a randomly empty cell. These stochastic disturbances add randomness and realism as reflecting external factors like job changes or lifestyle shifts that cause agents to relocate.

Modelling Questions

How does location preference influence clustering?

In the extended model, there are “desirable locations” which are preferred by the agents, such as parks or schools. This question evaluates whether agents having location preferences will speed up or hinder segregation, possibly overpower clustering by similarity, examining the relationship between location preferences and social preferences.

Can randomness disrupt segregation patterns?

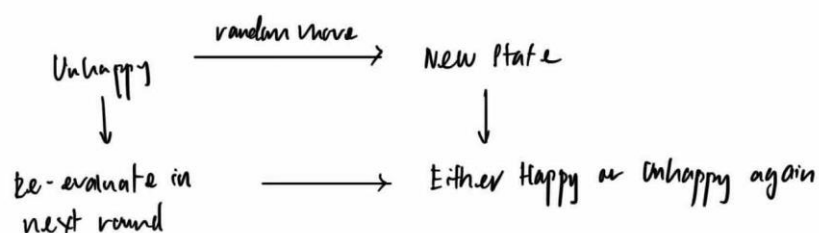
The model simulates unpredictable life occurrences by including a probability p of random relocation. This begs the question of whether even a small amount of stochastic movement is enough to disrupt established clusters of similar agents. This question evaluates whether stable segregation patterns are fragile or resilient in the face of external social or economic disruptions.

Techniques and Appropriateness

Markov Chain

Agent relocation in a Schelling-style segregation model is modelled in this simulation using a Markovian framework. It satisfied the key properties of a Markov process, although there is no formal transition matrix. The system will evolve in discrete time steps, and each agent's behaviour is dependent only on the present grid, not on any past states, making the process memoryless. The state of the agents is classified as happy or unhappy based on the proportion of neighbours of the same type. Monte Carlo stochasticity is introduced by randomly relocating unhappy agents to random empty cells. The probabilistic movement rules guarantee that transitions between states (location and satisfaction) are dependent only on the current state, making the model a discrete-time, stochastic Markov process.

State Diagram



Monte Carlo Simulation

By iteratively random sampling agent moves, it captures the emergent dynamics. In order to replicate the statistical behaviour over a large number of iterations, agents and empty cells are updated randomly. This is to evaluate steady-state segregation patterns under stochastic dynamics, accounting for the randomness and variety in agent updates.

Section 3: Model description

Model Extension Details

The extended Schelling model introduces location preferences and random relocations to the classic framework.

Location Preferences

Some grid cells are designated as “desirable location”. Agents evaluate their happiness based on neighbourhood similarity and distance to the “desirable location”. The closer an agent is to the “desirable location”, the happier they are, weighted by a parameter α .

Revised Happiness Satisfaction Rule

$$H_i = \left(\frac{\text{Similar Neighbors}}{\text{Total Neighbors}} \right) + \alpha \cdot \frac{1}{1 + d_i}$$

d_i : Manhattan distance to nearest desirable area

α : Weight controlling location importance

Random Relocations

Each agent has a given probability (p) of randomly moving to any unoccupied cell, regardless of their level of happiness, to simulate outside influences such as changes in lifestyle, family demands, or employment. The random move removes strictly deterministic move dynamics and adds stochastic noise to the system. This reflects the observation that people do move for reasons unrelated to local social dynamics.

List of Assumptions

Original Schelling Model

1. Binary groups: Two types of agents (e.g. Blue and Orange)
2. Agents are located on a 2D grid, with some empty cells to allow movement
3. Agents have a threshold preference that tolerates a minimum proportion of neighbours of the same type, else they will move if unhappy
4. Unhappy agents relocate to an empty cell to increase their happiness
5. Relocation is deterministic: No probabilistic decision making,
6. Static happiness threshold
7. No global preferences, only local interactions: No preference or bias for specific locations other than neighbour similarity

Extension Schelling Model

1. Binary groups: Two types of agents
2. Agents are located on a 2D grid, with some empty cells to allow movement
3. Agents have a threshold preference that tolerates a minimum proportion of neighbours of the same type, else they will move if unhappy
4. Unhappy agents relocate to an empty cell to increase their happiness
5. Happiness satisfaction dynamic includes location utility (α).
6. Stochasticity is introduced with random relocations: Agents may move randomly (probability *p*) or if unhappy
7. Markov property: transitions depend only on the current state (memoryless)
8. Global preferences introduced via distance to “desirable locations”
9. The process iterates until a steady state or fixed number of steps is reached

Model Class and Analysis

The discrete grid simulates agents of two types evaluating their happiness satisfaction level and proximity to the “desirable location”. Through random relocation probability p_{random} , the model adopts stochastic aspects, which enable both structured (location-aware) and unstructured (random) movement. Moreover, a simple average similarity index is used to quantify segregation as the system changes across discrete time steps. Due to the combination of local interactions, feedback loops, and randomness, the model is non-linear and stochastic, which is suitable for numerical simulation rather than analytical solution.

Algorithm and Mathematical Results or Derivations

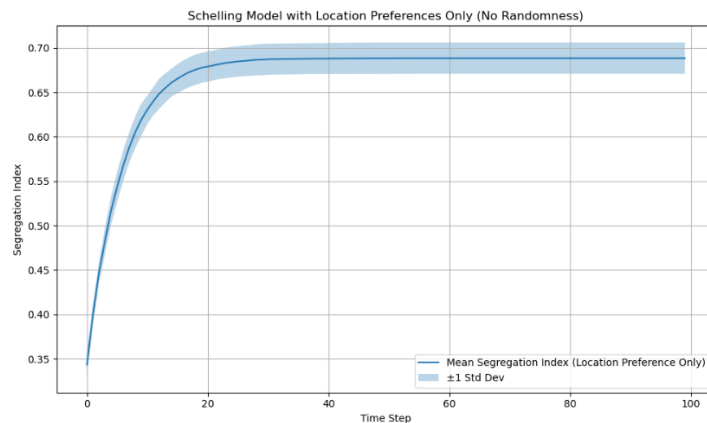
The extended Schelling model uses a 2D grid to simulate an agent-based stochastic relocation process. Each agent evaluates their happiness using the above function

$$H_i = \left(\frac{\text{Similar Neighbors}}{\text{Total Neighbors}} \right) + \alpha \cdot \frac{1}{1 + d_i}$$

Monte Carlo-style randomness is introduced when unhappy agents move to random empty cells when their happiness drops below a threshold as a result of stochastic shocks modelled by p_{random} probability. Each agent's average fraction of comparable neighbours is determined using the numerical segregation index, which tracks clustering behaviour over time. Key deviations include exogenous randomness, location preference via, and a quantified segregation metric.

Section 4: Results

Segregation Over Time Plot with Location Preferences (No Randomness)



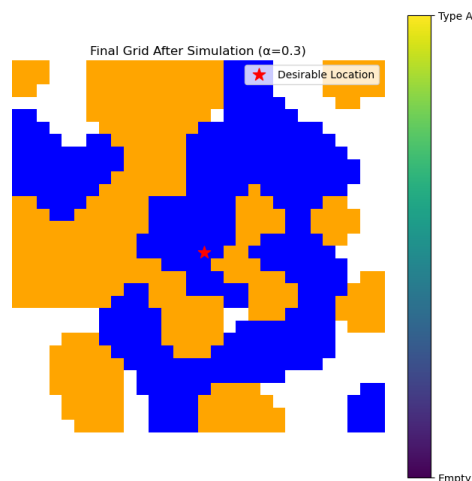
How did I arrive at the result?

I extended the model by adding the distance of each agent to the “desired location” to examine how location preferences affect segregation dynamics in Schelling's model. A weight, (α) is given to each agent to control how much they value being near the spot. If the agents are unhappy, they try to relocate to a better empty spot. A line plot is created to record the average neighbour similarity across the grid at each iteration over iterative time steps, which capture how segregation levels evolve.

Interpretation

The average similarity of an agent's neighbours is used by the model to track segregation; larger values denote more segregation. Within 30 steps, the index stabilises between 0.69 and 0.7 (high segregation level), having begun at roughly 0.34. The initial 20 steps' narrow shaded region (± 1 standard deviation) behaves consistently across the simulations. This demonstrates that deterministic location choices alone can cause significant segregation and agent cluster formation even in the absence of randomness evolving.

Final Grid Visualisation (without randomness)



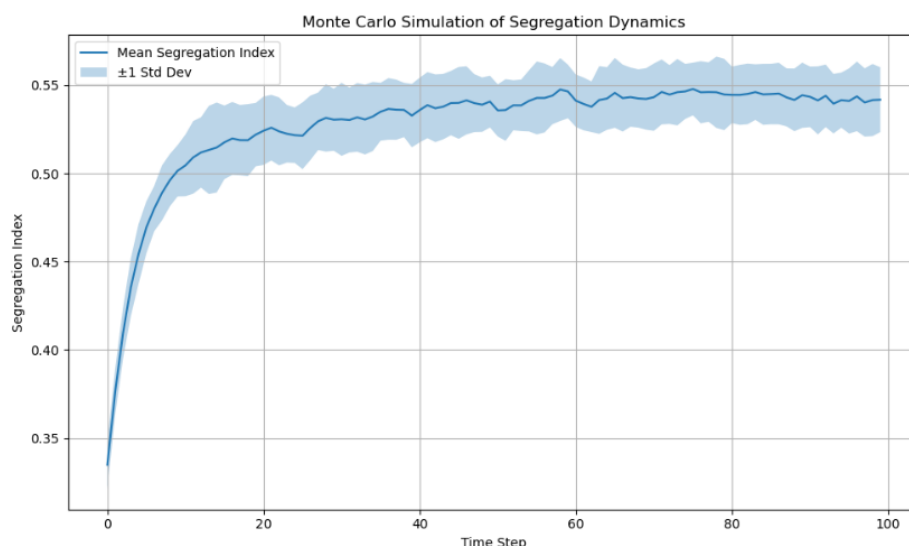
How did I arrive at the result?

I extended the Schelling model by adding a central “desire location” on a 30×30 grid, each agent's utility is combined with neighbourhood similarity and distance to centre ($\alpha=0.3$). Agents moved deterministically to empty cells only when the utility exceeded 0.3. The system stabilised when no more advantageous moves existed. This would result in many agents drawn toward the central area, indicating a balance between the preference for similar neighbours and the “desire location”.

Interpretation

The final grid heatmap shows how the system's appearance settled. Clusters of Type A (green) and Type B agents (yellow) are visible, along with a higher agent density close to the red star—the “desire location”. This demonstrates that the agents' preferred location did affect their movement. Because people want to be in good places and not just be near like-minded individuals which make the model more realistic.

Segregation Over Time Plot with Location Preferences (with Randomness)



How did I arrive at the result?

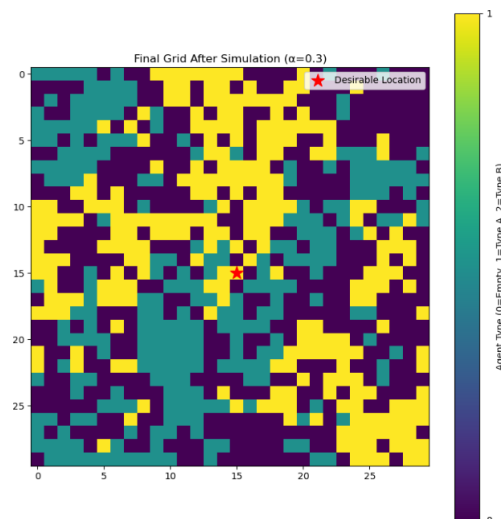
I initialise a 30×30 grid with 500 agents (250 per type) and 10% empty cells. The centre point (15,15) serves as the desirable location. Neighbourhood similarity and distance-based location preference are combined to determine the agent's happiness (weight=0.3). If an agent's happiness falls below 0.3 or with 5% random probability, they move. The simulation runs for 100 steps across 50 Monte Carlo trials, tracking the segregation index (mean $\pm$ standard deviation) over time.

Interpretation

The plot shows the segregation dynamics over 50 simulation trials, tracking the average fraction of similar neighbours per agent (segregation index) over 100 time steps. Segregation rises quickly as agents go to locations with stronger centrality (weight = 0.3) or more similar

neighbours after starting from random placement (index ~ 0.33). Growth of segregation slows down after step 20 stabilises between 0.54 and 0.55 as more similar neighbours reach satisfaction. Consistent results across trials are indicated by the narrow ± 1 SD band. Furthermore, the system reaches moderate clustering, which gives a glimpse of how location preferences and neighbours' similarity impact segregation. Moreover, the model doesn't fully segregate to 1, which suggests inherent randomness preventing it

Final Grid Visualisation (with randomness)



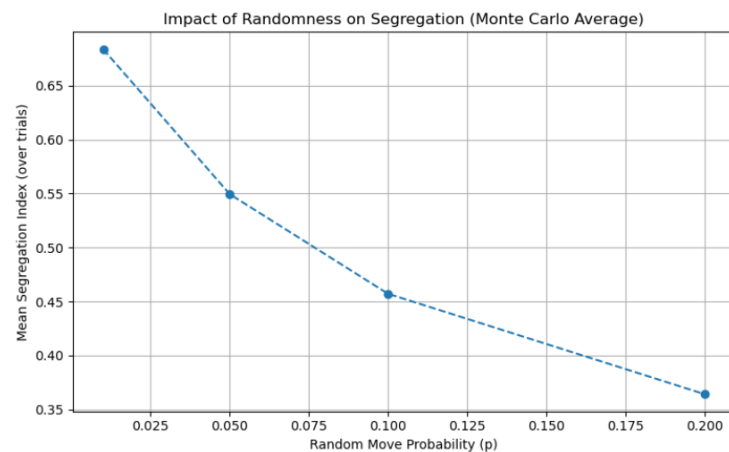
How did I arrive at the result?

The final grid state is visualised after the single simulation runs for 100 time steps with Type A (yellow), Type B (purple), empty (black) and desirable central location marked by red star at (15,15). Agents' movement rules are based on neighbours' similarity and location preference, leading to clear clustering of like agents while maintaining attraction toward the centre.

Interpretation

Although 5% random relocation and location preferences decrease segregation however they don't stop it. The segregation index continues to stabilise at 0.54-0.55, showing that even with stochastic elements, clustering is driven by agent preferences. The dynamics of segregation are not considerably disturbed by low-level stochastic relocation. As a result, segregation is a stable result when there is little stochastic movement since the underlying preference for similar neighbours and desirable locations dominates. Furthermore, agents may be able to avoid local suboptimal arrangements with the aid of a modest amount of randomness, which would subtly reinforce segregation.

Randomness and Segregation Disruption



How do I arrive at the result?

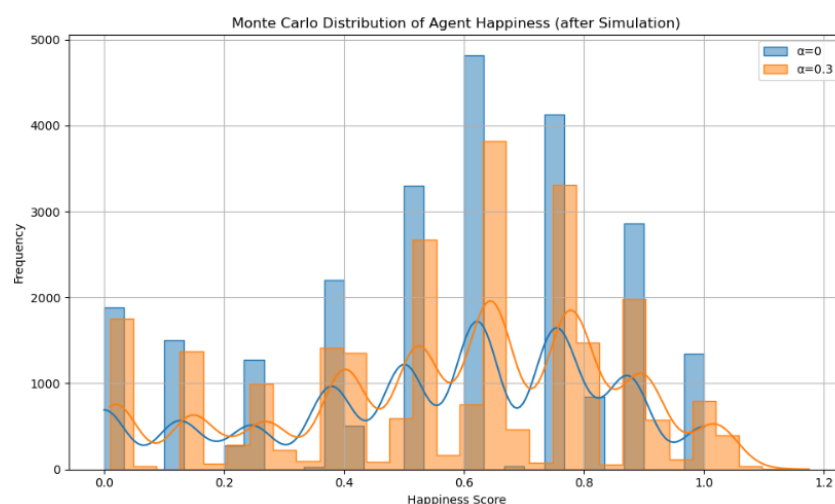
A Monte Carlo study of Schelling's model tested four randomness levels ($p = 0.01, 0.05, 0.10, 0.20$), with 20 trials each. Over 500 steps, agents moved if dissatisfied (by social/location thresholds) or randomly (with random move probability, p). Each trial's final segregation index was averaged per p to assess randomness' impact on emergent segregation.

Interpretation

The plot reveals an inverse relationship between randomness and segregation.

As p increases, the average level of segregation will decrease. However, this effect only becomes significant above a specific threshold: segregation clearly decreases when $p \geq 0.10$; low values of p (e.g., 0.01 or 0.05) have no effect. This suggests that random movement can disrupt stable, homogeneous clusters, but it should be high enough.

Happiness Distribution with/ without location preferences



How did I arrive at the result?

Two Schelling models are examined over 50 trials each for two different settings ($\alpha=0$ vs. $\alpha=0.3$). Agents moved depending on neighbourhood dissatisfaction or random chance. Location preference and neighbour similarity are used to calculate happiness satisfaction. Agents were only concerned with neighbourhood similarity when $\alpha = 0$, but when $\alpha = 0.3$, pleasure was influenced by both social and location preferences. Aggregated results were visualised via KDE-enhanced histograms, showing how location preference impacts happiness distributions.

Interpretation

Key differences in agent happiness between $\alpha=0$ (blue) and $\alpha=0.3$ (orange) are displayed in the histogram. A clustered distribution results when $\alpha=0$, meaning that happiness is solely dependent on neighbours. The curve swings to the right at $\alpha=0.3$, indicating greater levels of overall pleasure as location preferences help make up for less optimal social settings. Inequality in the advantages of location preference is revealed by the fact that certain agents are still unhappy ($H_i < 0.3$) if they are unable to access preferred locations.

Conclusion of Results

Answer to Question 1

Adding location preferences will cause agents to care not just about neighbours' similarity but also location preferences. In the original model, agents move only based on neighbours' similarity, which creates strong segregation over time. But now, introducing location preferences will cause agents to be more concentrated around the "desired location" and the pattern becomes more realistic.

Answer to Question 2

Stochastic relocation disrupts emergent segregation patterns only when the probability of random movement is sufficiently high (e.g., $p \geq 0.10$). At low levels of randomness, random moves are too infrequent to significantly alter the agents' tendency to cluster with similar neighbours, and thus segregation remains robust. However, as the randomness increases, the regularity of disruptive movement interferes with the formation and stability of homogeneous clusters, leading to a noticeable reduction in segregation.

Section 5: List of algorithms and concepts

Algorithms:

1. Agent-Based Simulation – Grid-based agent movement
2. Monte Carlo Simulation – Random sampling for stochastic behaviour.
3. Markov Chain Process – Memoryless state transitions.
4. Happiness Calculation – Combines neighbour similarity and distance to desirable locations.

5. Random Relocation – Fixed probability p of forced moves to simulate lifestyle changes
6. Segregation Index – Measures clustering over time.

Key Concepts:

1. Schelling's Segregation Model – Original neighbour-based movement.
2. Location Preference (α) – Agents choose regions that are close to their “desired location”
3. Stochastic Relocation (p) – Clustering is disrupted by random movements
4. Emergent Behaviour – Large scale patterns arise from simple interactions between agents following local rules
5. Manhattan Distance – Measures spatial utility.
6. Nonlinear Dynamics – Agent decision making feedback loops
7. Steady-State Analysis – Convergence of segregation levels.
8. Threshold-Based Movement – Agents relocate if unhappy ($H_i < \text{threshold}$).

AI Statement

I acknowledge the use of ChatGPT (<https://chat.openai.com/>)

i) To refine my own work and academic language

I submitted the analysis I wrote for each plot with instructions to improve my phrasing, vocabulary, and language accuracy (4 iterations). I further rephrased to represent my tone and style of writing.

ii) To help clarify the concepts related to the Schelling model, Markov Property, Manhattan Distance, Stochastic Relocation and Emergent Behaviour

I asked ChatGPT about the differences between my extension model and the original Schelling model in this related concept. I ask about the further details related to the concept. (10 iterations)